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$$r = 2 \cos \theta$$



$$\begin{aligned} L &= 2 \int_0^{\frac{\pi}{2}} \sqrt{(2 \cos \theta)^2 + ((2 \cos \theta)')^2} d\theta \\ &= \int \sqrt{(2 \cos \theta)^2 + ((2 \cos \theta)')^2} d\theta \\ &= \int \sqrt{4 \cos^2 \theta + 4 \sin^2 \theta} d\theta \\ &= 2 \int \sqrt{1} d\theta \\ &= 2\theta + c \\ \Rightarrow L &= 2 [2\theta]_0^{\frac{\pi}{2}} = 2\pi \end{aligned}$$

References